

LATENT REWARD REGISTERS FOR DIFFUSION PREFERENCE ALIGNMENT

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ABSTRACT

Aligning diffusion models with human preferences usually relies on a sparse terminal reward evaluated on the final generated samples, presenting a severe temporal credit-assignment challenge across the multi-step denoising process. We propose *Latent Reward Registers*, a mechanism that estimates terminal preference directly from intermediate noisy latents by prepending learnable, position-free register tokens to the input sequence of a frozen Diffusion Transformer (DiT). This independent readout mechanism extracts latent reward evidence without altering the generator’s hidden states or velocity field. The resulting dense, differentiable reward signal throughout the full denoising process facilitates two alignment strategies. For training, *Reward-Gradient On-Policy Distillation* (RG-OPD) distills reward-guided updates along on-policy trajectories, bypassing the computationally expensive rollouts of standard policy gradients. For inference, *Reward-Guided Sampling* (RGS) steers trajectories via magnitude-matched reward gradients without parameter updates. Empirically, at high noise levels ($u=0.8$), the registers reach the highest pairwise accuracy among the evaluated latent reward models. Furthermore, RG-OPD outperforms online reinforcement learning baselines while reducing GPU hours by up to $33\times$, and RGS establishes a new state-of-the-art among training-free methods, strictly enhancing both alignment and perceptual metrics. Code and weights are available at <https://github.com/Guanys-dar/latent-reward-register>

1 INTRODUCTION

While diffusion and flow-matching models have achieved remarkable success in generating high-fidelity visual content, aligning these models with human preferences remains a critical challenge. Current methods generally rely on reward maximization (Lee et al., 2023; Clark et al., 2024; Liu et al., 2025; Zheng et al., 2026). An external reward model evaluates the final generated image, and the generator is updated to maximize this scalar score, thereby improving perceptual quality and prompt alignment.

Existing alignment methods typically fall into two paradigms. *Reward backpropagation* (Lee et al., 2023; Wu et al., 2024; Clark et al., 2024; Prabhudesai et al., 2023) unrolls the sampling trajectory to compute exact gradients from a differentiable reward model. While this enables fast optimization, it often leads to mode collapse. Alternatively, *online reinforcement learning (RL)* (Black et al., 2023; Liu et al., 2025; Xue et al., 2025; Zheng et al., 2026) formulates generation as a sequential policy optimized via policy gradients. This mitigates mode collapse but introduces high variance and low sample efficiency.

Both paradigms face the same bottleneck: they rely on a single scalar reward evaluated only on the fully generated image. Because this terminal score cannot be easily attributed to specific intermediate steps, the supervision is severely delayed. To pass this delayed signal back, reward backpropagation must unroll long generation chains in memory, making it prone to over-optimization. Online RL, on the other hand, requires a large number of samples to keep gradient variance under

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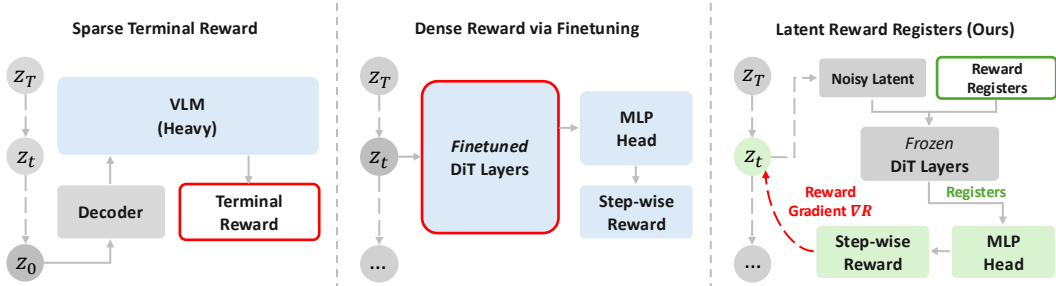


Figure 1: **Latent Reward Registers vs. existing paradigms.** Unlike conventional alignment methods that rely on delayed, sparse feedback from a clean image (Left) or require intrusive fine-tuning of the generator (Middle), our approach (Right) appends learnable registers to a frozen DiT. This enables the direct estimation of step-wise rewards from intermediate noisy latents (z_t). Consequently, we derive dense, differentiable reward gradients (∇R) for generation while keeping the pretrained generative dynamics fully intact.

control. Furthermore, adapting the generator to a novel reward function requires full retraining, as these terminal objectives offer no straightforward mechanism for test-time guidance.

Our central insight is that DiT internal representations already encode sufficient evidence about the final output quality, well before the trajectory reaches a fully denoised state. Even under heavy noise, these features capture emerging global structure, prompt alignment, and overall trajectory viability. Rather than propagating a single endpoint score backward through a full rollout, we can therefore estimate the expected terminal preference directly from each intermediate latent, turning sparse terminal feedback into a dense, step-wise reward signal without requiring complex approximations.

Building on this insight, we introduce *Latent Reward Registers*, a non-intrusive mechanism that turns a frozen DiT into an accurate, step-wise reward model. As illustrated in Fig. 1, a small set of learnable, position-free register tokens is added as global readouts to aggregate reward-relevant evidence from the intermediate representations of the frozen transformer. A lightweight readout head then predicts the expected terminal preference directly from any intermediate latent. By differentiating this prediction with respect to the latent, we obtain a local reward gradient at every denoising step. Because the registers do not modify the generator’s hidden states or velocity predictions, the pretrained generative dynamics remain fully intact. As a result, the framework provides dense, differentiable reward supervision with minimal computation overhead.

This dense reward signal supports alignment during both training and inference. For training, *Reward-Gradient On-Policy Distillation* (RG-OPD) distills reward-guided updates along on-policy trajectories, bypassing the computationally expensive rollouts of standard policy gradients. For inference, *Reward-Guided Sampling* (RGS) steers trajectories via magnitude-matched reward gradients without parameter updates, allowing multiple objectives to be composed at test time. Our contributions are summarized below.

- **Reliable latent reward signals.** Latent Reward Registers estimate step-wise rewards directly from intermediate noisy latents. Even at high noise levels, their pairwise ranking accuracy remains competitive with external reward models that require fully denoised images.
- **Efficient training-time alignment.** By distilling register-guided on-policy trajectories, RG-OPD surpasses online RL baselines in alignment quality while reducing GPU hours by up to $33\times$.
- **Effective inference-time alignment.** RGS steers trajectories using the derived latent reward gradients without parameter updates, improving target rewards while maintaining a superior reward-quality trade-off relative to existing training-free sampling methods.

2 RELATED WORK

Diffusion Preference Alignment. Preference alignment for text-to-image generators is dominated by training-time methods. Reward backpropagation (Lee et al., 2023; Clark et al., 2024; Prabhudesai et al., 2023) differentiates through the denoising chain to optimize the generator directly, while on-

line reinforcement learning (Liu et al., 2025; Zheng et al., 2026) treats generation as a policy refined with policy gradients. A fundamental bottleneck for these methods is their reliance on sparse end-point rewards, which severely complicates credit assignment across intermediate denoising steps. To bypass this sparsity, recent on-policy distillation (OPD) frameworks (???) distill step-wise signals from task-specialized teachers or context-conditioned self-teacher along the student’s own rollouts. However, scaling these pipelines to multi-objective alignment requires independently pre-training a separate RL teacher for each target reward. Querying these full-sized teacher networks at every denoising step introduces heavy computational overhead during training, and inherently makes inference-time alignment infeasible. Training-free alignment offers an inference-time alternative, but existing schemes still suffer from high computational costs, whether through iterative search and optimization (Yeh et al., 2025; Tang et al., 2025) or guidance gradients that become unstable at high noise levels (Song et al., 2023). Our approach instead extracts a dense, differentiable step-wise reward directly in the latent space, keeping alignment lightweight and efficient at both training and inference time.

Latent Reward Models. Most reward models operate on clean, decoded images (Xu et al., 2023; Kirstain et al., 2023; Ma et al., 2025). Applying them to intermediate states would require forming a clean estimate from each intermediate latent via expensive VAE decoding at every step. Moreover, the resulting signals are noisy or unreliable on heavily corrupted inputs. Recent latent reward models instead evaluate noisy states directly, drawing evidence from selected internal features (Zhang et al., 2025), early cross-attention maps (Cui et al., 2026), or separate process reward models (Mi et al., 2025; Go et al., 2026; Yang et al., 2024). Existing methods typically rely on heuristic feature engineering or computationally heavy auxiliary networks. Latent Reward Registers instead append lightweight, position-free register tokens to a frozen DiT, establishing a differentiable reward mechanism that avoids manual feature selection and leaves the base generator unaltered. This design directly supports both RG-OPD and RGS.

Test-time Guidance. Guidance methods steer a pretrained generator without retraining it. Classifier guidance injects the gradient of an auxiliary classifier into the score estimate (Dhariwal & Nichol, 2021), and classifier-free guidance (CFG) extrapolates between conditional and unconditional predictions to strengthen prompt adherence (Ho & Salimans, 2022); the latter is standard in modern text-to-image diffusion. RGS continues this lineage: it leaves the generator and its native text guidance untouched and appends a reward-gradient correction whose magnitude is matched to the sampler’s own displacement. Because the correction is derived from latent registers at intermediate steps, sampling needs neither image decoding nor a separate reward-model backbone.

3 METHOD

To address the temporal credit-assignment problem in preference alignment, we construct a dense, differentiable reward signal directly from intermediate latent states, providing a unified supervision mechanism for both training and inference time alignment. The proposed framework consists of three core components:

- **Latent Reward Registers** (Sec. 3.1): A lightweight mechanism that predicts the expected terminal preference using learnable register tokens and frozen DiT features. This auxiliary stream extracts step-wise reward signals from intermediate noisy latents without perturbing the original velocity field (Fig. 2).
- **Reward-Gradient On-Policy Distillation** (Sec. 3.2): A training-time alignment strategy that constructs detached one-step optimization targets along the student’s sampling trajectory. This bypasses backpropagation through complete denoising chains and eliminates the high variance of rollout-level policy gradients.
- **Reward-Guided Sampling** (Sec. 3.3): An inference-time alignment technique that steers latent trajectories using magnitude-matched, step-wise reward gradients, requiring no additional parameter updates.

3.1 REWARD REGISTER MECHANISM

Motivation. Conventional image reward models supervise only the terminal generated image, and applying them to intermediate states would require repeated decoding that exposes the reward signal

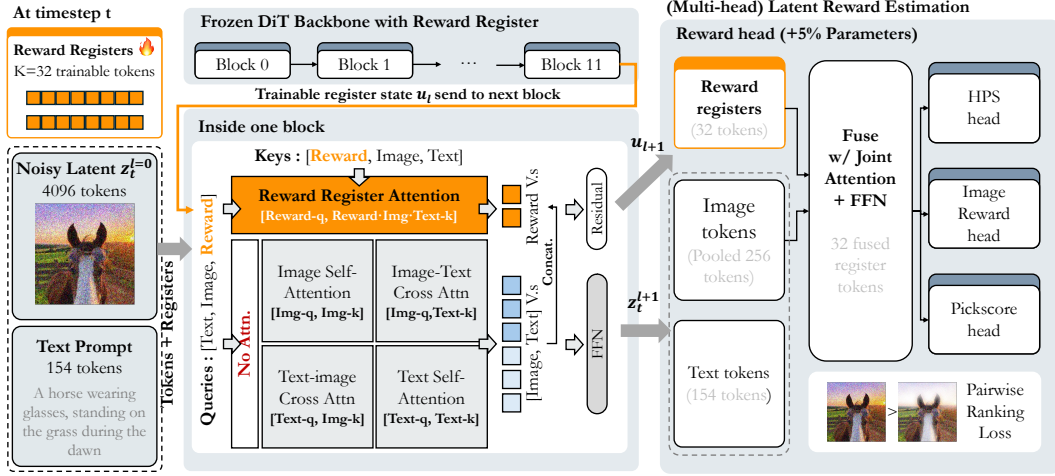


Figure 2: **Overview of the Reward Register mechanism.** Learnable registers establish a non-intrusive auxiliary stream through a frozen DiT. These registers derive representations from frozen image and text features without altering the original token sequence, thereby strictly preserving the original velocity field. A lightweight fusion module then combines the final register states with intermediate backbone representations for multi-head latent reward estimation under pairwise ranking supervision.

to noisy artifacts. DiT representations, in contrast, encode prompt alignment, layout, and appearance early in the generation process. We therefore introduce a lightweight side-stream readout that predicts terminal preference directly from a noisy latent z_i . Unlike a static feature probe, the register state persists across multiple frozen DiT blocks, accumulating evidence across depth. Because it derives representations from the backbone without altering the spatial token sequence, the side stream provides differentiable step-wise supervision while preserving the original velocity field.

Notation. Let z_σ denote a latent state at continuous noise level $\sigma \in [0, 1]$, with sampling proceeding from $\sigma_{\max}$ to 0. During reward-register training and evaluation, we use u interchangeably with σ to denote the same normalized noise level, and paired clean latents z_{clean} are forward-noised as $z_u = (1 - u)z_{\text{clean}} + u\epsilon$. For a numerical schedule $\sigma_0 > \dots > \sigma_N = 0$, we write $z_i := z_{\sigma_i}$.

Non-intrusive reward registers. We initialize K learnable, position-free register tokens $Q^{(0)} \in \mathbb{R}^{K \times d}$ as global readouts (Darcet et al., 2024). For each latent state, we set $Q_i^{(0)} = Q^{(0)}$. Let $H_i^{(l)} = [H_{i,\text{img}}^{(l)}, H_{i,\text{txt}}^{(l)}]$ denote the native image and text hidden states at block l for latent z_i . For the initial L transformer blocks, the registers share a single trainable query projection W_Q^ϕ while reusing the frozen key, value, output, normalization, and noise-level-gating modules from each block. Letting $\bar{Q}_i^{(l)}$, $\bar{H}_{i,\text{img}}^{(l)}$, and $\bar{H}_{i,\text{txt}}^{(l)}$ denote the σ_i -conditioned normalized states, the update is:

$$\begin{aligned} K_i^{(l)} &= [\bar{Q}_i^{(l)} W_K^{(l)}, \bar{H}_{i,\text{img}}^{(l)} W_K^{(l)}, \bar{H}_{i,\text{txt}}^{(l)} W_{K,\text{txt}}^{(l)}], \\ V_i^{(l)} &= [\bar{Q}_i^{(l)} W_V^{(l)}, \bar{H}_{i,\text{img}}^{(l)} W_V^{(l)}, \bar{H}_{i,\text{txt}}^{(l)} W_{V,\text{txt}}^{(l)}], \\ Q_i^{(l+1)} &= Q_i^{(l)} + \gamma_i^{(l)} \odot \text{Attn}(\bar{Q}_i^{(l)} W_Q^\phi, K_i^{(l)}, V_i^{(l)}) W_O^{(l)}, \end{aligned} \quad (1)$$

where $\gamma_i^{(l)}$ is the frozen backbone’s σ_i -conditioned attention gate. This design isolates the side stream: native image and text streams are computed independently of the registers, and for single-stream architectures (e.g., FLUX.1-dev) the native sequence serves directly as the combined input context. The registers also bypass the block feed-forward networks, so the original denoiser hidden states and velocity predictions stay strictly invariant. Finally, the register states are fused with pooled frozen features from selected blocks through a lightweight attention module, producing shared per-register features $U_i \in \mathbb{R}^{K \times d}$ for the reward heads.

Algorithm 1 Reward-Gradient On-Policy Distillation (RG-OPD). Detached one-step reward-tilted targets are constructed at states visited by the current student.

```

1: Input: prompt distribution  $\mathcal{C}$ ; frozen reference field  $v^{\text{ref}}$ ; frozen reward register  $\{r_\phi^{(m)}\}$ ; sampler
   steps  $\{\mathcal{S}_i\}_{i=0}^{N-1}$ ; guidance schedule  $\{\alpha_i\}_{i=0}^{N-1}$ ; optimized steps  $\mathcal{J}$ 
2: Output: reward-aligned student field  $v_\theta$ 
3: Initialize  $v_\theta \leftarrow v^{\text{ref}}$ 
4: for each training round do
5:   Sample prompts  $c \sim \mathcal{C}$ 
6:   Sample a student trajectory  $\{z_i\}_{i=0}^N$  using  $v_\theta$  and set  $\bar{z}_i \leftarrow z_i$ 
7:   for  $i = 0, \dots, N-1$  do
8:     Compute  $\mu_i^{\text{ref}} \leftarrow \mathcal{S}_i(\bar{z}_i; v^{\text{ref}})$ 
9:     if  $\alpha_i > 0$  then
10:      Compute the detached single- or multi-reward direction  $g_i$ 
11:    else
12:      Set  $g_i \leftarrow 0$ 
13:    end if
14:    Compute and store the teacher target  $\mu_i^{\text{RG}}$  using Eq. 4
15:  end for
16:  Shuffle the stored trajectories into training minibatches
17:  for  $i \in \mathcal{J}$  do
18:    Evaluate the student mean  $\mathcal{S}_i(\bar{z}_i; v_\theta)$ 
19:    Accumulate Eq. 5 and update only  $\theta$ 
20:  end for
21: end for

```

Reward prediction and learning. For reward objective m , the latent score is

$$r_\phi^{(m)}(z_i, c, \sigma_i) = \frac{1}{K} \sum_{k=1}^K h_\phi^{(m)}(U_{i,k}), \quad m = 1, \dots, M, \quad (2)$$

where $h_\phi^{(m)}$ is a reward-specific two-layer MLP. For a pair (a, b) , let $s_{ab}^{(m)}$ be the sign of the score difference assigned by endpoint reward model m , and let $d_{ab}^{(m)} = r_\phi^{(m)}(z_{u,a}, c, u) - r_\phi^{(m)}(z_{u,b}, c, u)$. Each head is trained with the noise-aware Thurstone pairwise-ranking objective (Thurstone, 1927) together with the DiNa-LRM variance adjustment (Liu et al., 2026): the loss penalizes disagreement between the signs of $d_{ab}^{(m)}$ and $s_{ab}^{(m)}$, and the assumed comparison variance grows with the sampled noise level. Images within a prompt group share the same noise level and Gaussian noise realization, which controls corruption as a pairwise confound while preserving differences between the underlying clean images. The backbone stays frozen; only the registers, shared query projection, fusion readout, and reward heads are optimized, and an exponential moving average of the parameters is used for evaluation and downstream use. The learned score preserves the rankings induced by the endpoint reward models rather than estimating a calibrated reward value.

Shared reward direction. Both RG-OPD and RGS utilize gradients derived from the learned latent scores. For a single reward objective m , the gradient is $g^{(m)}(z) = \nabla_z r_\phi^{(m)}(z, c, \sigma)$. For multi-reward optimization we normalize each gradient to unit root-mean-square (RMS) and sum them, so that variations in gradient magnitude across heads do not dominate the aggregate direction. We define $\text{unit}_{\text{RMS}}(x) := x / \max(\text{RMS}(x), \delta_{\text{rms}})$, with RMS computed across all latent dimensions per sample and $\delta_{\text{rms}} > 0$ for numerical stability. The resulting single- or multi-reward direction $g(z)$ is detached before being used in either an RG-OPD target or an RGS update.

3.2 REWARD-GRADIENT ON-POLICY DISTILLATION

Reward-Gradient On-Policy Distillation (RG-OPD) generates step-wise supervision at states actively visited by the *current student model*, adapting the training distribution to the evolving student rather than relying on fixed offline teacher trajectories. This mechanism performs local one-step dis-

Algorithm 2 Reward-Guided Sampling. A frozen reward register adds magnitude-matched reward-gradient corrections to an otherwise unchanged reference sampler.

```

1: Input: prompt  $c$ ; frozen reference field  $v^{\text{ref}}$ ; frozen reward register  $\{r_\phi^{(m)}\}$ ; sampler steps  $\{\mathcal{S}_i\}_{i=0}^{N-1}$ ; noise levels  $\{\sigma_i\}_{i=0}^N$ ; guidance schedule  $\{\alpha_i\}_{i=0}^{N-1}$ ; selected heads  $\mathcal{M}$ 
2: Output: generated image  $x$ 
3: Initialize  $z_0 \sim \mathcal{N}(0, I)$  and set schedule  $\sigma_0 > \dots > \sigma_N = 0$ 
4: for  $i = 0, \dots, N - 1$  do
5:   Compute  $\mu_i^{\text{ref}} \leftarrow \mathcal{S}_i(z_i; v^{\text{ref}})$  and  $\Delta z_i^{\text{ref}} \leftarrow \mu_i^{\text{ref}} - z_i$ 
6:   if  $\alpha_i > 0$  then
7:     Compute and detach the direction  $g_i$  aggregated over the selected heads  $\mathcal{M}$ 
8:     Compute  $z_{i+1}$  via Eq. 7
9:   else
10:    Set  $z_{i+1} \leftarrow \mu_i^{\text{ref}}$ 
11:   end if
12: end for
13: Decode  $z_N$  to obtain  $x$ 

```

tillation, entirely bypassing backpropagation through the full sampling chain and avoiding rollout-level policy-gradient estimation.

Let $\mathcal{S}_i(z; v)$ denote the deterministic one-step sampler mean associated with the transition from σ_i to σ_{i+1} under velocity field v . From a student rollout $\{z_i\}_{i=0}^N$, we define the detached state $\bar{z}_i = \text{sg}(z_i)$ and evaluate the frozen reference generator to obtain

$$\mu_i^{\text{ref}} = \mathcal{S}_i(\bar{z}_i; v^{\text{ref}}), \quad \Delta z_i^{\text{ref}} = \mu_i^{\text{ref}} - \bar{z}_i. \quad (3)$$

Evaluating the shared reward direction at the detached state gives $g_i = g(\bar{z}_i)$. The reward-tilted teacher mean is

$$\mu_i^{\text{RG}} = \mu_i^{\text{ref}} + \alpha_i \text{RMS}(\Delta z_i^{\text{ref}}) \cdot \text{unit}_{\text{RMS}}(g_i). \quad (4)$$

The coefficient α_i controls the relative scale of the reward correction with respect to the reference displacement. It is zero on unguided steps, for which the target reduces to the frozen reference mean and acts as an explicit retention constraint.

For deterministic flow-matching trajectories, we optimize the mean squared error between the student’s one-step mean and the detached teacher target. Let D be the number of scalar latent dimensions and $\mathcal{J} \subseteq \{0, \dots, N - 1\}$ the configured set of optimized denoising steps. Then

$$\mathcal{L}_{\text{RG-OPD}}(\theta) = \frac{1}{|\mathcal{J}|D} \sum_{i \in \mathcal{J}} \|\mathcal{S}_i(\bar{z}_i; v_\theta) - \text{sg}(\mu_i^{\text{RG}})\|_2^2. \quad (5)$$

The reference generator and reward register remain frozen, and gradients update only the student parameters θ . Algorithm 1 summarizes the procedure; backbone-specific parameterizations and the choice of $\mathcal{J}$ are implementation details reported in Appendix B.

3.3 REWARD-GUIDED SAMPLING

Reward-Guided Sampling (RGS) utilizes the learned latent reward field to steer the generation process at inference time. With the generator and the register both frozen, RGS uses the original sampler as a baseline reference and applies a reward-gradient correction after each active solver step. Let $\mathcal{S}_i$ denote the deterministic solver transition from σ_i to σ_{i+1} , incorporating any native text guidance. The reference mean and displacement are given by:

$$\mu_i^{\text{ref}} = \mathcal{S}_i(z_i; v^{\text{ref}}), \quad \Delta z_i^{\text{ref}} = \mu_i^{\text{ref}} - z_i. \quad (6)$$

For an active step, the register yields the detached single- or multi-reward direction $g_i = g(z_i)$. The trajectory advances according to:

$$z_{i+1} = \mu_i^{\text{ref}} + \alpha_i \text{RMS}(\Delta z_i^{\text{ref}}) \text{unit}_{\text{RMS}}(g_i), \quad (7)$$

where α_i controls the relative scale of the reward correction with respect to the reference displacement. Because Eq. 7 depends on the displacement of a complete reference-solver step, it composes

Table 1: **High-noise pairwise preference accuracy (%) on four benchmarks.** Latent reward models score noisy states at $u=0.8$, whereas endpoint reward models score clean images. **Red bold** and **blue underline** mark the best and second-best results within each model category.

Model	Backbone	Pairwise Preference Accuracy (%)				
		ImageReward	HPDv2	HPDv3	GenAI-Bench	Avg
<i>Endpoint Reward Models</i>						
ImageReward	CLIP	65.15	73.95	58.74	63.41	65.31
PickScore	CLIP	62.73	79.44	65.67	69.98	69.45
HPSv2	CLIP	65.62	82.58	64.69	67.62	70.13
HPSv3	Qwen2VL-7B	67.03	85.36	76.03	70.95	74.84
<i>Latent Reward Models</i>						
LRM-SD1.5	SD-1.5	55.88	64.80	52.37	52.46	56.38
LRM-SDXL	SDXL	58.13	66.49	52.87	54.67	58.04
PAVRM	SD3-Medium	56.76	73.52	68.18	57.20	63.91
Diffusion Probe	SD3-Medium	58.67	75.29	68.68	57.14	64.95
DiNa-LRM	SD3-Medium	60.10	78.76	68.08	58.94	66.47
Reward Register	SD3-Medium					
+HPS		61.84	78.46	69.96	61.41	67.92
+ImageReward		63.32	75.73	63.20	62.61	66.22
+Raw score-sum		63.12	78.10	69.10	62.95	68.32

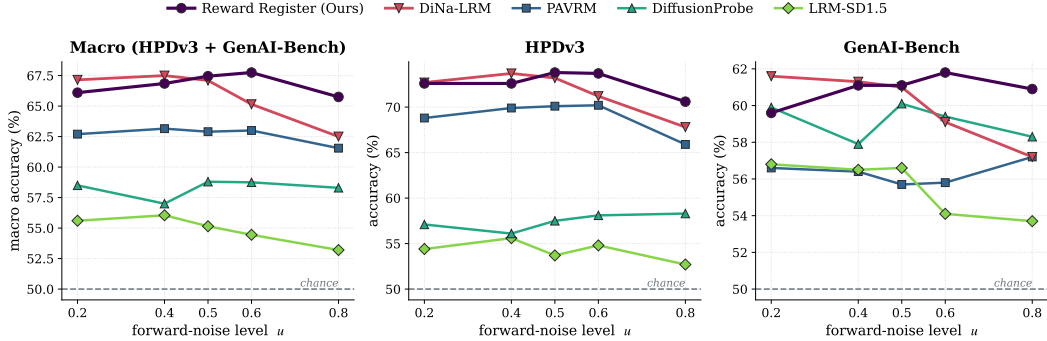


Figure 3: **Preference accuracy across noise levels.** Pairwise preference accuracy of latent reward models scored at noise level u .

with deterministic solvers of any order, and the reward correction is applied exactly once after that step.

Applying corrections at low noise levels brings marginal benefit and can distort fine details, so we use a three-band schedule: $\alpha_i = \alpha_{\text{early}}$ for $\sigma_i > \sigma_{\text{hi}}$, $\alpha_i = \alpha_{\text{mid}}$ for $\sigma_{\text{lo}} < \sigma_i \leq \sigma_{\text{hi}}$, and $\alpha_i = 0$ otherwise, with $\alpha_{\text{early}} > \alpha_{\text{mid}}$. Algorithm 2 outlines RGS; backbone-specific sampling configurations are detailed in Appendix B.

4 EXPERIMENTS

We evaluate the framework along three axes: prediction accuracy of Latent Reward Registers on noisy latents, training efficiency of RG-OPD against RL and reward backpropagation baselines, and inference effectiveness of RGS relative to training-free sampling methods. Primary evaluations use SD3-Medium, with FLUX.1-dev providing cross-architecture evidence; we then ablate register designs and analyze noise robustness. Full protocols and extended analyses appear in the Appendix.

Table 2: **Training-time preference alignment on SD3-Medium and FLUX.1-dev.** HPSv3 and ImageReward are optimization targets, MUSIQ and CLIP-IQA measure no-reference perceptual quality. **Red bold** and **blue underline** mark the best and second-best results within each category.

Method	SD3-Medium				FLUX.1-dev			
	HPSv3↑	IR↑	MUSIQ↑	CIQA↑	HPSv3↑	IR↑	MUSIQ↑	CIQA↑
Baselines								
Base + CFG	8.55	1.183	73.2	0.694	9.89	1.142	72.9	0.669
ReFL +HPS	10.84	1.341	67.0	0.512	12.00	1.337	56.8	0.428
ReFL +IR	9.69	1.678	75.0	<u>0.716</u>	9.31	1.293	62.7	0.425
AlignProp +HPS	11.20	1.472	68.1	0.580	11.83	1.241	59.4	0.481
AlignProp +IR	9.51	<u>1.611</u>	71.2	0.491	9.47	1.213	68.2	0.577
Flow-GRPO +HPS	10.26	1.349	74.3	0.692	<u>12.06</u>	1.212	75.9	0.697
Flow-GRPO +IR	9.45	1.297	74.1	0.705	11.13	1.377	<u>75.7</u>	<u>0.702</u>
Diffusion-NFT +HPS	9.51	0.995	72.4	0.551	11.54	1.056	75.1	0.695
Diffusion-NFT +IR	9.68	1.221	74.4	0.686	11.50	1.241	74.1	0.699
Reward Register with on-policy distillation (Ours)								
Reward Register (OPD) +HPS	<u>11.57</u>	1.336	<u>74.9</u>	0.726	12.38	1.229	74.4	0.648
Reward Register (OPD) +IR	10.98	1.444	72.1	0.674	10.82	<u>1.367</u>	75.6	0.704
Reward Register (OPD) +MH	11.62	1.417	74.0	0.704	11.91	1.346	75.0	0.681

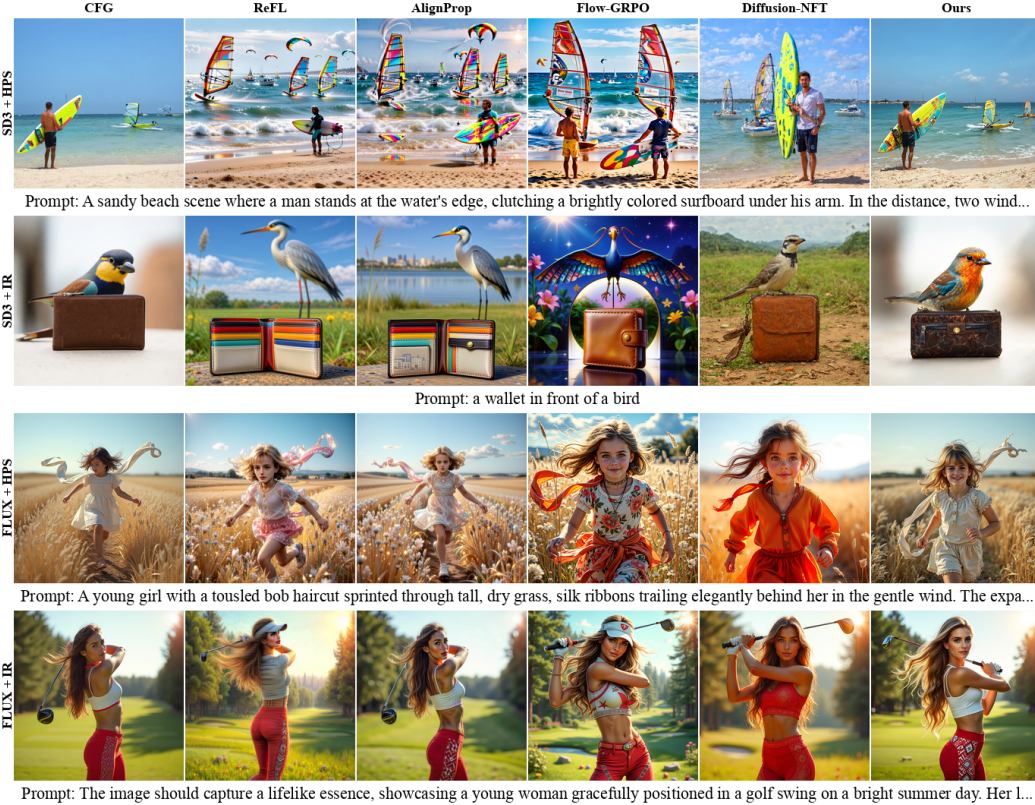


Figure 4: **Qualitative comparison of training-time alignment.** We compare the visual results of our RG-OPD against baseline methods, including CFG, ReFL, AlignProp, Flow-GRPO, and Diffusion-NFT. The generated samples demonstrate that our approach achieves better prompt alignment and higher perceptual quality without introducing visible artifacts.

Table 3: **Reward-Guided Sampling with frozen SD3-Medium and FLUX.1-dev generators.** RGS provides the strongest HPS alignment and a favorable reward–quality trade-off among the evaluated training-free methods. HPSv3 and ImageReward (IR) are optimization targets; MUSIQ and CLIP-IQA (CIQA) measure perceptual quality. DNO and Demon use HPSv3 for their +HPS variants. **Red bold** and **blue underline** mark the best and second-best results within each backbone.

Method	SD3-Medium				FLUX.1-dev			
	HPSv3↑	IR↑	MUSIQ↑	CIQA↑	HPSv3↑	IR↑	MUSIQ↑	CIQA↑
Baselines								
Base + CFG	8.55	1.183	73.2	0.694	9.89	1.142	72.9	0.669
DNO +HPS	8.85	1.272	73.1	0.698	9.64	1.191	71.9	0.655
Demon +HPS	9.09	1.311	73.0	0.685	9.56	1.209	68.2	0.605
DNO +IR	8.19	1.454	72.3	0.693	9.25	<u>1.350</u>	69.5	0.632
Demon +IR	8.90	1.596	72.5	0.674	9.27	1.501	68.2	0.603
Reward-Guided Sampling (Ours)								
Reward Register +HPS	10.70	1.347	75.0	0.728	11.03	1.212	<u>73.3</u>	<u>0.715</u>
Reward Register +IR	10.12	<u>1.465</u>	73.0	0.694	10.38	1.297	73.2	0.710
Reward Register +MH	<u>10.48</u>	1.413	<u>74.1</u>	<u>0.723</u>	<u>10.80</u>	1.276	73.5	0.717

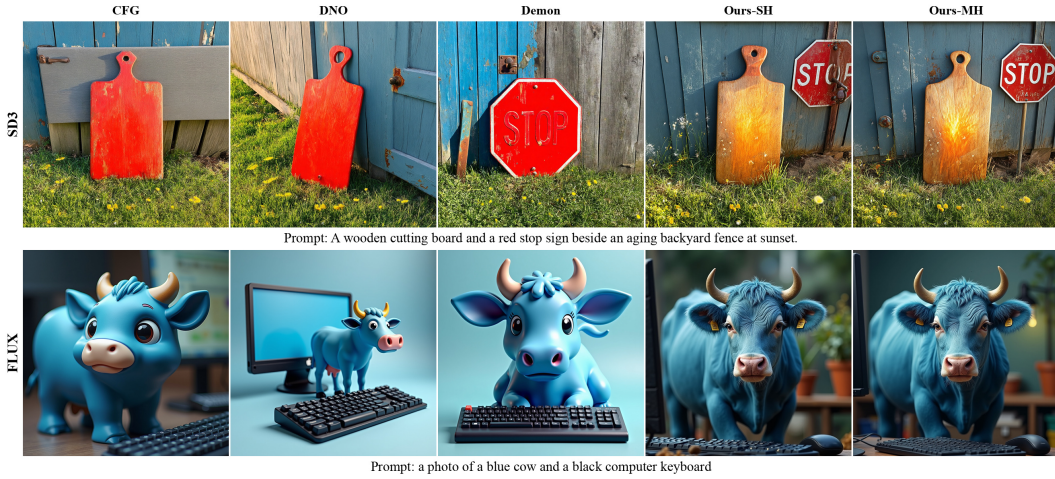


Figure 5: **Comparison of training-free sampling methods.** CFG, DNO, Demon, and our single- and multi-head RGS variants use matched prompts and seeds within each backbone.

4.1 LATENT REWARD PREDICTION

Setup. We append $K=32$ register tokens to a frozen SD3-Medium backbone, training only the auxiliary branch. We measure pairwise preference accuracy across four benchmarks totaling 54,170 pairs: ImageReward (Xu et al., 2023), HPDv2 (Wu et al., 2023), HPDv3 (Ma et al., 2025), and GenAI-Bench (Li et al., 2024). Accuracy is defined as the frequency with which a model assigns a higher score to the human-preferred image within a held-out pair. Latent reward models, including our method, evaluate paired latents forward-noised to the high-noise regime $u=0.8$. Standard endpoint models (ImageReward, PickScore, HPSv2, HPSv3) score the clean decoded images, providing a cross-regime reference. We compare against five latent reward baselines: LRM (Zhang et al., 2025) on SD1.5 and SDXL, PAVRM (Mi et al., 2025)(re-implemented on SD3-Medium), Diffusion Probe (Cui et al., 2026), and DiNa-LRM (Liu et al., 2026). For fair comparison, Table 1 matches our backbone to the SD3-Medium baselines; Appendix C details FLUX.1-dev results.

Results. The reward register predicts preferences accurately from noisy inputs: it outperforms all latent baselines on three of the four benchmarks (Table 1) and remains highly competitive on the

Table 4: **Reward Register Design Ablation.** Pairwise preference accuracy evaluated across four benchmarks. Δ quantifies the deviation in average accuracy relative to the full deployed method. **Bold** indicates the optimal performance within each respective column.

Variant	Pairwise Preference Accuracy (%)					Δ
	ImageReward	HPDv2	HPDv3	GenAI-Bench	Avg	
Full method	63.12	78.10	69.10	62.95	68.32	—
w/o reward register	56.86	74.12	65.42	60.76	64.29	−4.03
w/o skip-FFN	58.15	77.82	68.54	60.38	66.23	−2.09
w/o trainable q-proj	58.20	77.62	68.62	60.42	66.21	−2.11
w/ layers [4]	59.39	79.32	69.57	61.69	67.49	−0.83
w/ layers [4, 8]	60.01	79.90	70.35	61.30	67.89	−0.43

fourth, despite evaluating heavily corrupted latents. Clean-image reward models score uncorrupted inputs and are included as a reference point rather than a matched comparison. Appendix C shows that these prediction capabilities transfer to FLUX.1-dev.

4.2 REWARD-GRADIENT ON-POLICY DISTILLATION

Setup. We benchmark RG-OPD against reward-backpropagation (ReFL, AlignProp) and online RL methods (Flow-GRPO, Diffusion-NFT). All approaches update the generator weights. Evaluations utilize matched prompts, seeds, resolutions, and metric pipelines. HPSv3 and ImageReward serve as the optimization targets, while MUSIQ and CLIP-IQA assess reward-independent perceptual quality. Appendix B details the complete training protocol.

Results. RG-OPD attains the highest HPSv3 scores on both architectures while keeping a balanced multi-reward profile (Table 2). Reward-backpropagation baselines, by contrast, often maximize target metrics at the expense of visible perceptual degradation, a classic signature of reward over-optimization. In contrast, RG-OPD achieves a superior Pareto frontier between reward alignment and perceptual quality. Figure 4 shows this qualitatively: under matched prompts and seeds, RG-OPD preserves prompt-relevant content and perceptual quality without the artifacts exhibited by several baselines.

4.3 REWARD-GUIDED SAMPLING

Setup. We compare RGS with the training-free baselines DNO and Demon under matched prompts, seeds, resolutions, and metrics. All methods keep the generator frozen; DNO and Demon optimize or search per sample using external reward models, whereas RGS uses analytical gradients from the latent reward register at each active step. HPSv3 and ImageReward quantify target alignment, and MUSIQ (Ke et al., 2021) and CLIP-IQA (Wang et al., 2023) measure no-reference perceptual quality. Extended protocol details are given in Appendix B.

Results. Each single-head RGS variant improves its target metric relative to the CFG baseline while preserving perceptual quality, and RGS reaches the highest HPSv3 score among all evaluated training-free methods, with a favorable reward–quality trade-off under ImageReward optimization. Figure 5 gives the corresponding qualitative comparison; prompts and seeds are matched to hold input variation fixed, strictly isolating the observed improvements to the proposed guidance mechanism rather than prompt or seed selection.

Beyond alignment accuracy, RGS demonstrates a distinct memory efficiency advantage. Because it shares the frozen denoiser backbone, peak memory stays comparable to unguided CFG and well below that of guided baselines, which require separate external scorer networks (Table 10).

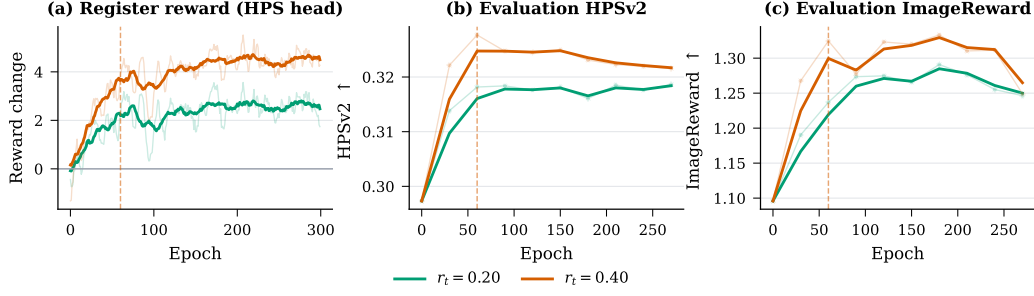


Figure 6: **Effect of reward tilt (r_t) during RG-OPD training.** Training dynamics are compared between $r_t=0.20$ and $r_t=0.40$. The results show that a larger reward scale accelerates convergence and yields a higher peak performance. Moreover, the upward trend of the internal latent reward (a) remains consistent with the endpoint HPSv2 evaluation (b), indicating alignment between the intermediate proxy and the terminal objective. ImageReward (c) is evaluated as an independent, unoptimized metric to monitor the overall generation quality during the process.

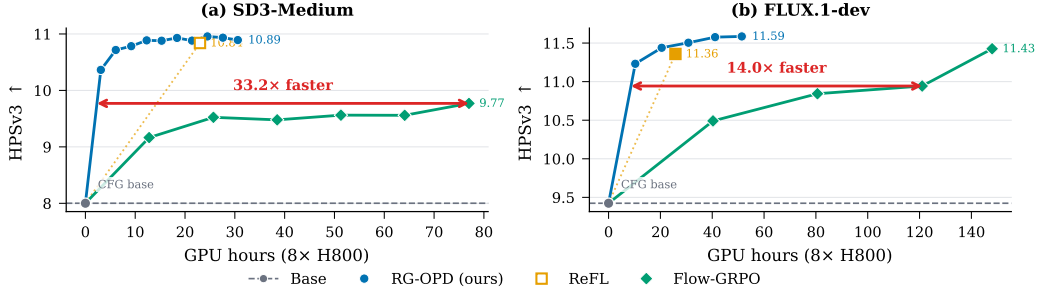


Figure 7: **Training efficiency of RG-OPD.** We compare the HPSv3 optimization trajectories of RG-OPD and Flow-GRPO against aggregate GPU hours on SD3-Medium and FLUX.1-dev. The curves indicate that RG-OPD requires fewer GPU hours to reach equivalent reward thresholds. This acceleration comes from the dense, step-wise gradients extracted from the latent reward registers.

4.4 ANALYSIS

Register design. We observe that maintaining a persistent register state is essential for accurate step-wise reward estimation. As shown in Table 4, removing the register tokens leads to the most severe performance degradation, while the skip-FFN connection and trainable query projection yield complementary improvements. Interestingly, reducing the depth of readout layers has a marginal impact. This implies that the bottleneck for latent reward prediction lies in accumulating temporal context via the side-stream register, rather than relying on heuristic layer selection.

Training efficiency. Compared to standard policy-gradient RL, RG-OPD benefits from dense, step-wise gradient feedback during the denoising process. This formulation bypasses the high variance typical of sequence-level trajectory sampling, accelerating convergence significantly. As shown in Figure 7, RG-OPD achieves Flow-GRPO-equivalent HPSv3 scores $14.0\times$ – $33.2\times$ faster in GPU hours across both backbones, while on-policy rollouts ensure the proxy model remains aligned with the shifting student distribution. Note that to track training dynamics at high frequency, the efficiency curves are evaluated on a 256-prompt validation subset, whereas the final benchmarks (Table 2) strictly report on the full 400/800 test set to ensure rigorous evaluation.

Multi-head RGS. RGS naturally extends to multi-objective optimization without requiring complex hyperparameter tuning. While single-head guidance is inherently restricted to a specific reward and prone to over-optimization, the multi-head variant effectively trades off conflicting objectives, preserving both reward gains and perceptual quality (Table 3, Figure 8). The equal-RMS aggrega-

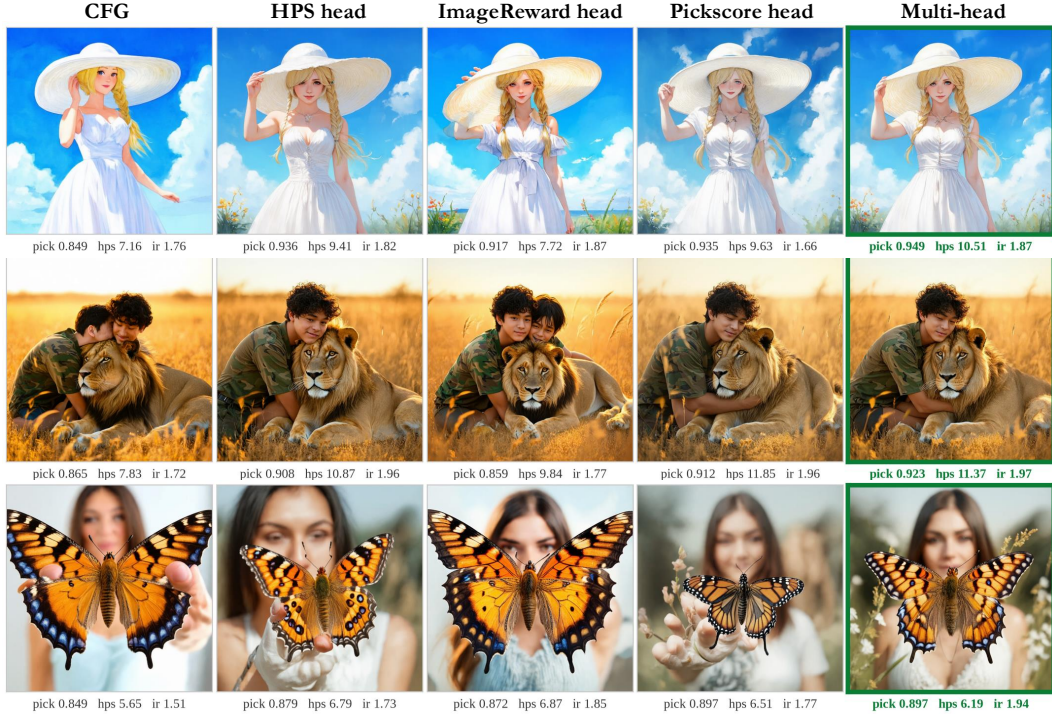


Figure 8: **Multi-head RGS.** Multi-head reward gradients achieve a balanced performance across different reward metrics, compared to single-head RGS under identical settings.

tion strategy is crucial here: it standardizes the gradient magnitudes across different reward heads, preventing any single objective from dominating the guidance direction at a given timestep.

Robustness in high-noise regimes. A key advantage of the register mechanism is its stability at high noise levels. While baselines like DiNa-LRM perform competitively near the data distribution ($t \rightarrow 0$), their accuracy drops sharply when latent inputs become predominantly noise. In contrast, the register maintains reliable prediction capabilities across the entire timestep range (Figure 3). This property is particularly critical for early-stage guidance ($t \rightarrow T$), where the latent trajectory dictates the global structure and semantic layout of the final image.

Additional analysis. Appendix E demonstrates the robustness of our approach under varying sampling budgets, guidance scales, and ODE solvers, alongside an analysis of how reward corrections accumulate over the sampling trajectory. Furthermore, Appendix C verifies that individual register heads successfully preserve the relative ranking distributions of their corresponding endpoint reward models. These results indicate that the step-wise proxy avoids local mode collapse, providing reliable and generalizable feedback regardless of the specific inference configuration.

5 CONCLUSION

In this work, we demonstrate that intermediate DiT representations capture terminal reward information early in the denoising process. Building on this insight, we propose Latent Reward Registers, a non-intrusive auxiliary stream that predicts endpoint rewards directly from intermediate noisy latents. The resulting dense reward field supports two alignment strategies. Reward-Gradient On-Policy Distillation (RG-OPD) utilizes step-wise gradient supervision to outperform online RL baselines in efficiency and alignment, while avoiding the over-optimization typical of direct reward backpropagation. Furthermore, Reward-Guided Sampling (RGS) applies magnitude-matched register gradients to steer generation without parameter updates, improving multi-objective target rewards while preserving perceptual quality. Ultimately, by resolving the credit assignment problem, this framework establishes a reliable latent reward model for diffusion RL.

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A APPENDIX OVERVIEW

This appendix contains supplementary material: the experimental protocols, extended results for Latent Reward Registers, training-time alignment, and Reward-Guided Sampling (RGS), and directions for future research.

1. **Experimental Settings** (Appendix B)
 - Reward register training (Appendix B)
 - Reward-gradient on-policy distillation (Appendix B)
 - Reward-Guided Sampling (Appendix B)
2. **Additional Reward-Register Results** (Appendix C)
 - Pairwise preference accuracy (Appendix C.1)
 - Rank agreement with endpoint reward models (Appendix C.2)
3. **Additional Training-Time Alignment Results** (Appendix D)
 - Qualitative results (Appendix D.1)
4. **Additional Reward-Guided Sampling Results** (Appendix E)
 - Magnitude-matched CFG-direction control (Appendix E.1)
 - Inference cost (Appendix E.2)
 - Sampling budgets and guidance strengths (Appendix E.3)
 - Reward along the guided trajectory (Appendix E.4)
 - Robustness to the ODE solver (Appendix E.5)
 - Qualitative comparison across prompts (Appendix E.6)
5. **Future Work** (Appendix F)

B EXPERIMENTAL SETTINGS

Reward Register Training Setting Table 6 lists the hyperparameters and configuration details for training the reward register that were omitted from the main text, distinguishing backbone-specific architectural choices from the shared optimization settings.

For the evaluation on the HPDv3 benchmark, 28 designated test pairs that exhibited data overlap with the SD3 training corpus were excluded. The remaining evaluation benchmarks detailed in Table 1 constitute independent validation sets.

Reward-Gradient On-Policy Distillation Setting Table 5 lists the optimization configurations and hyperparameters for SD3-Medium and FLUX.1-dev. Direct epoch-to-epoch comparisons between RG-OPD and the online RL baselines would be misleading, since the two paradigms consume data at different rates; the main text therefore reports computational efficiency in aggregate GPU hours.

Table 5: **RG-OPD optimization details.** Model checkpoints were selected using a validation set disjoint from the final test set. Computational budgets are reported in terms of GPU hours and optimizer updates to facilitate fair comparison across paradigms with varying rollout densities.

Setting	SD3-Medium Configuration	FLUX.1-dev Configuration
Student adaptation	LoRA; merged for evaluation	LoRA; merged for evaluation
Training length	60–100 epochs	150 epochs
Training rollout	10 FlowMatch Euler steps	10 FlowMatch Euler steps
Optimized denoising steps $\mathcal{J}$	First 9 of 10 rollout steps ($\rho = 0.99$)	First 9 of 10 rollout steps ($\rho = 0.99$)
Teacher reward tilt α_i	{0.20, 0.40, 0.50} for $\sigma_i > 0.2$; 0 otherwise	0.80 for $\sigma_i > 0.2$; 0 otherwise
Flow-GRPO / Diffusion-NFT	LoRA $r=32/\alpha=64$; 6-step SDE; 100 epochs	LoRA $r=32/\alpha=64$; 6-step SDE; 100 epochs
Rollout samples (RL baselines)	1,152 rollout images per epoch	1,152 rollout images per epoch

Table 6: **Reward-register training details.** This table outlines the backbone-specific configuration and shared training hyperparameters. Values spanning both columns indicate shared settings across architectures.

Setting	SD3-Medium	FLUX.1-dev
Register-equipped blocks L	12 (of 38)	8 (of 19 double blocks)
Feature-snapshot blocks $\mathcal{L}$	{4, 8, 12}	{9, 19, 28}
Register tokens K	32	
Image-token pooling	64×64 to 16×16	
Readout attention heads	8	
Endpoint reward objectives	HPSv3, ImageReward	
Training data	30,652 prompt groups, 4 images/group	
Validation data	300 prompt groups	
Data split	98/1/1 by prompt	
Training length	3 epochs	
Batch size	64	
Precision / optimizer	bf16 AdamW	
Learning rate	5×10^{-5}	
Weight decay	0.01	
Warmup / schedule	1,000 steps / constant	
Gradient clipping	1.0	
EMA decay	0.999	

Table 7: **Reward-Guided Sampling details.** Both SD3-Medium and FLUX.1-dev employ identical noise interval bands, with backbone-specific sampling step counts and mid-band guidance magnitudes.

Setting	SD3-Medium	FLUX.1-dev
Resolution	1024×1024	1024×1024
Sampler	FlowMatch Euler	FlowMatch Euler
Sampling steps	42	40
Text guidance	CFG scale 4.5	Embedded guidance 3.5
Early band ($\sigma_i > 0.8$)	$\alpha_{\text{early}} = 0.30$	$\alpha_{\text{early}} = 0.30$
Mid band ($0.2 < \sigma_i \leq 0.8$)	$\alpha_{\text{mid}} = 0.05$	$\alpha_{\text{mid}} = 0.10$
Low-noise tail ($\sigma_i \leq 0.2$)	0	0
Multi-head combination	Equal-weight sum of unit-RMS per-head gradients	
Correction mode	Additive; no spatial gradient filtering	

Reward-Guided Sampling Setting Table 7 documents the sampling configurations and guidance hyperparameters omitted from the main text.

C ADDITIONAL REWARD-REGISTER RESULTS

C.1 PAIRWISE PREFERENCE ACCURACY

Pairwise preference accuracy measures how often a model scores the human-preferred image higher within a held-out pair. The evaluation spans the ImageReward, HPDv2, HPDv3, and GenAI-Bench datasets, with 54,170 pairs in total; the HPDv3 set is restricted to 14,372 pairs after 28 samples are excluded for overlap with the training data. Endpoint reward models are evaluated on fully decoded, pristine images, while latent reward models score pairs that have been forward-noised to $u=0.8$. Across both backbones, +Raw score-sum denotes an unweighted ensemble of the constituent register heads.

On FLUX.1-dev the raw score-sum ensemble reaches an average accuracy of 70.00% (Table 8), confirming that the register mechanism transfers to an alternative frozen backbone.

Table 8: **FLUX.1-dev high-noise pairwise preference accuracy (%)**. The register mechanism, coupled with a frozen FLUX.1-dev backbone, is systematically evaluated at a noise level of $u=0.8$, adhering strictly to the evaluation protocols and benchmark datasets established in Table 1.

Variant	ImageReward	HPDv2	HPDv3	GenAI-Bench	Avg
+HPS	62.12	81.18	73.30	61.75	69.59
+ImageReward	63.13	78.84	65.86	63.42	67.81
+Raw score-sum	63.46	81.20	72.36	62.96	70.00

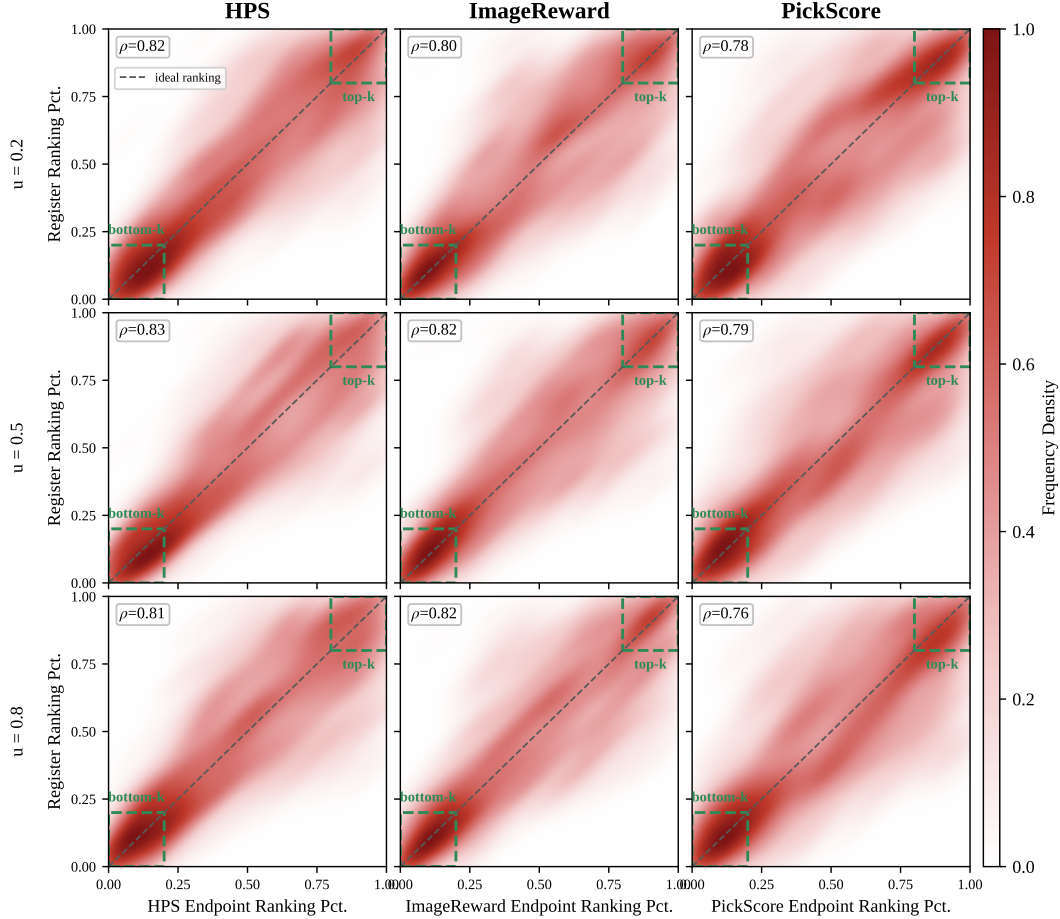


Figure 9: **Rank agreement between register heads and endpoint reward models**. The figure displays the joint probability density of ranking percentiles derived from register-head evaluations on noised latents, contrasted with corresponding percentiles obtained from endpoint models on fully decoded images. The parameter ρ denotes the Spearman rank correlation coefficient, and the diagonal line represents theoretically perfect rank preservation.

C.2 RANK AGREEMENT WITH ENDPOINT REWARD MODELS

Each register head preserves the ranking behavior of its corresponding endpoint reward model at high noise. Figure 9 compares the ranking percentiles the register assigns to noised latents with those of the endpoint models on fully denoised images, over a sample of 800 images. The Spearman correlation ρ ranges from 0.76 to 0.83 and stays stable across noise levels $u \in \{0.2, 0.5, 0.8\}$.

D ADDITIONAL TRAINING-TIME ALIGNMENT RESULTS

D.1 QUALITATIVE RESULTS

Figure 10 provides additional qualitative examples of training-time alignment. Each row uses the same text prompt and initial latent seed, so the comparison reflects consistent gains in prompt adherence and visual quality across prompts.

E ADDITIONAL REWARD-GUIDED SAMPLING RESULTS

E.1 MAGNITUDE-MATCHED CFG-DIRECTION CONTROL

To separate the effect of correction magnitude from correction direction, the CFG Control baseline replaces the RGS reward gradient with the SD3 conditional-unconditional residual. At each active step the applied correction is $\alpha_i \text{RMS}(\Delta z_i^{\text{ref}}) \text{unit}_{\text{RMS}}((\sigma_{i+1} - \sigma_i)(v_{\text{cond}} - v_{\text{uncond}}))$, following the same α_i schedule as RGS. The two controls therefore share the per-step magnitude-matching rule and differ only in the direction applied.

CFG Control improves some perceptual and reward metrics relative to standard CFG but does not approach the alignment of RGS (Table 9). Since the correction magnitude is identical, this comparison isolates the learned reward direction as the driver of the improvements.

Table 9: **Magnitude-matched direction control on SD3-Medium.** The CFG Control mechanism is formulated to exactly match the per-step root-mean-square (RMS) magnitude of our proposed correction, while strictly orienting the update vector along the standard CFG residual rather than the learned reward direction. All evaluations adhere to the fixed prompt and seed protocol defined in Table 3. **Red bold** and **blue underline** denote the highest and second-highest performance metrics within each respective category.

Method	HPSv3 $\uparrow$	IR $\uparrow$	MUSIQ $\uparrow$	CIQA $\uparrow$
Base + CFG	8.55	1.183	73.2	0.694
CFG Control	8.50	1.214	<u>74.14</u>	0.720
Reward Register +HPS	10.70	1.347	75.01	0.728
Reward Register +IR	10.12	1.465	73.00	0.694
Reward Register +MH	<u>10.48</u>	<u>1.413</u>	74.10	<u>0.723</u>

E.2 INFERENCE COST OF REWARD-GUIDED SAMPLING

Reward Register has a memory footprint comparable to standard CFG and an inference latency competitive with the most efficient guided baselines (Table 10). All reported numbers are measured end-to-end on a single uncontended H800 GPU in bf16, generating 1024^2 images over 42 steps with CFG scale 4.5 and batch size 1. Each method runs in its own process and is averaged over ten runs after ten warm-up runs. In the standard deployment the register shares the frozen denoiser and text encoders of the sampler, so the memory overhead is limited to the 147.6 M trainable parameters of the read path and reward head: peak memory rises only from 16.9 GiB to 17.6 GiB versus unguided CFG (a $1.04\times$ factor). Baselines that require separate neural backbones instead consume 27–34 GiB. Computationally, each guided step adds roughly 50 ms and 5.9 TFLOPs on top of the 12.4 TFLOPs of a base generation step.

End-to-end latency is $1.80\times$ that of standard CFG, slightly above PAVRM ($1.67\times$) but far below Diffusion Probe ($2.63\times$) and the decode-then-ImageReward pipeline ($4.12\times$). Restricting guidance to the early phase ($\sigma_i > 0.8$, i.e., 18 of the 42 steps) lowers the overhead to $1.34\times$ CFG while keeping roughly 80% of the HPSv3 gains of the full scheduler.

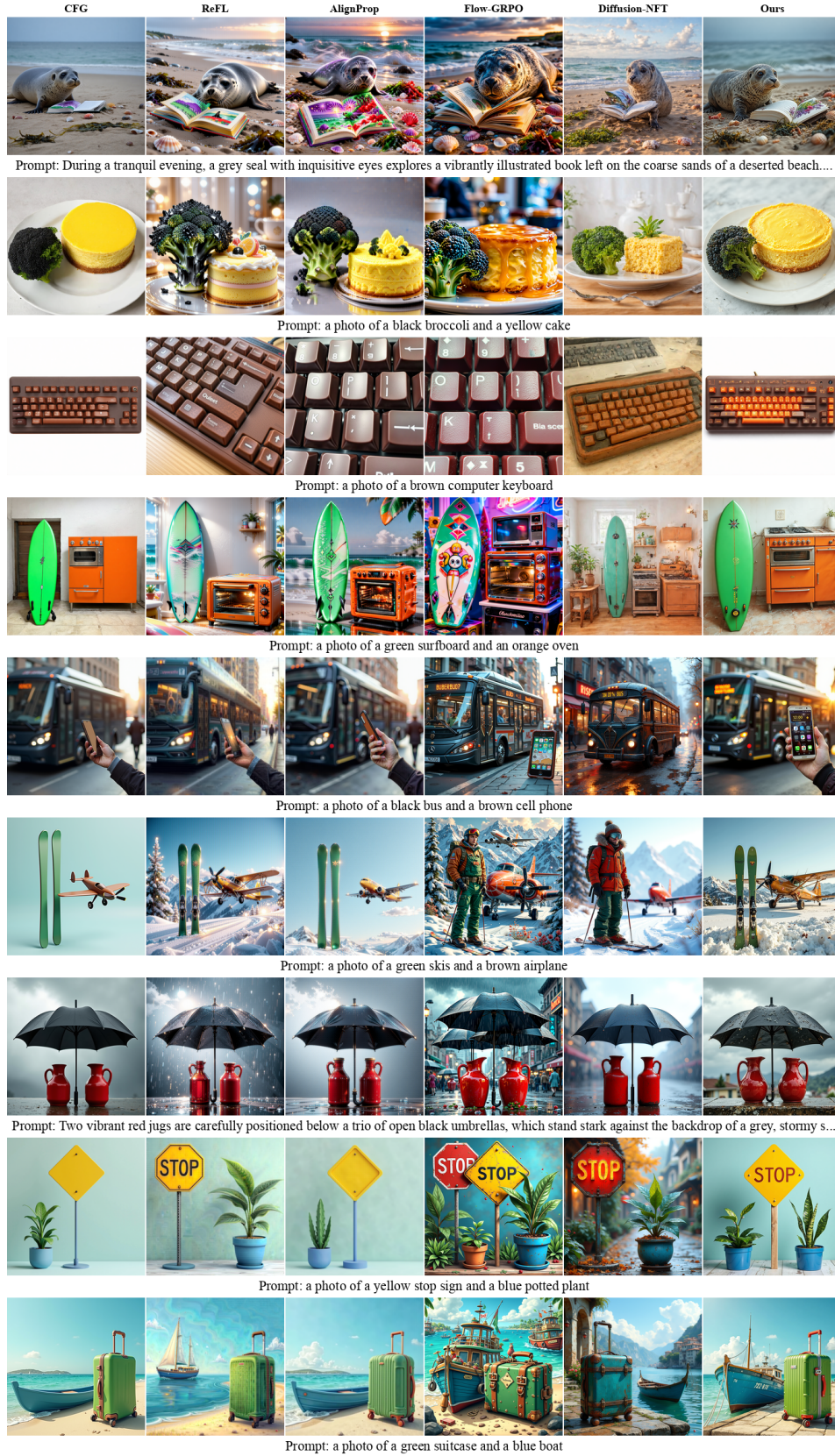


Figure 10: **Extended qualitative comparison of training-time alignment.** We provide additional visual examples comparing our RG-OPD framework against baseline methods across a diverse set of text prompts.

Table 10: **Inference cost of RGS on SD3-Medium.** Comparison of trainable parameter budgets, memory footprints, and computational overhead against existing guided baselines. SD3 with standard CFG is provided as the unguided reference. $\Delta\text{FLOP}/\text{step}$ denotes the marginal computational overhead relative to a base CFG step. The *early-only* configuration applies guidance exclusively during the first 18 of 42 steps (where $\sigma_i > 0.8$). **Bold** indicates the most efficient performance among guided methods (lower is better).

Method	Params↓ (M)	Peak mem↓ (GiB)	Reward Eval./step↓ (ms)	$\Delta\text{FLOP}/\text{step}↓$ (TFLOP)	Latency/img↓ (s, $\times\text{CFG}$)
<i>Reference (no reward)</i>					
SD3 + CFG	—	16.9	—	12.4 (base)	2.99 (1.00 $\times$)
<i>Guided baselines (separate reward backbone)</i>					
PAVRM	691.4	31.3	40	+4.1	4.98 (1.67 $\times$)
Diffusion Probe	15.3	34.3	108	+11.5	7.85 (2.63 $\times$)
Decode+ImageReward	446.7	27.3	215	+10.2	12.32 (4.12 $\times$)
<i>Reward Register (frozen generator, reward register)</i>					
Reward Register +HPS	147.6	17.6	50	+5.9	5.38 (1.80 $\times$)
Reward Register +HPS (<i>early-only</i>)	147.6	17.6	50	+5.9	4.00 (1.34$\times$)

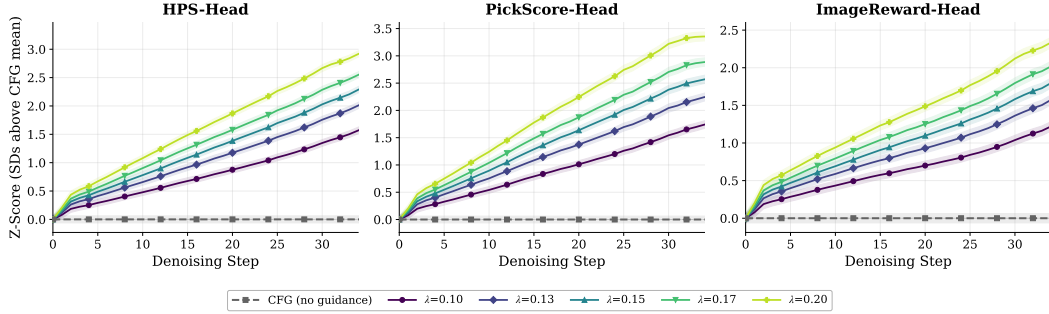


Figure 11: **Reward progression during generation under uniform guidance.** Register-head evaluations are plotted as z -scores, normalized against the per-step distribution of the unguided CFG baseline. Curves denote different guidance strengths α (labeled as λ in the legend), with shaded regions indicating variance across diverse text prompts.

E.3 RGS ACROSS SAMPLING BUDGETS AND STRENGTHS

Figure 12 evaluates uniform guidance strengths under sampling budgets of 10, 20, 30, and 40 steps, in contrast to the two-band scheduler used in the main deployment (42 steps). Guidance improves every metric relative to the CFG baseline at all budgets, and the targeted reward metrics generally rise with guidance strength. No-reference metrics such as CLIP-IQA and MUSIQ also exceed the CFG baseline but plateau, or vary only marginally, at higher strengths. The parameter λ in the figure legends is mathematically equivalent to the guidance strength α defined in the method section.

E.4 REWARD ALONG THE RGS TRAJECTORY

The reward-gradient correction exhibits progressive accumulation along the generation trajectory, proportionally scaling with the specified guidance magnitude. This auxiliary analysis uses a uniform guidance strength $\alpha \in [0.10, 0.20]$ rather than the two-band scheduler of the main deployment. Figure 11 tracks the register-head scores for HPSv3, PickScore, and ImageReward through the denoising process. The correction raises the scores by 2–3.3 standard deviations relative to the CFG baseline mean, and the ordering of guidance strengths is preserved across all three evaluators.

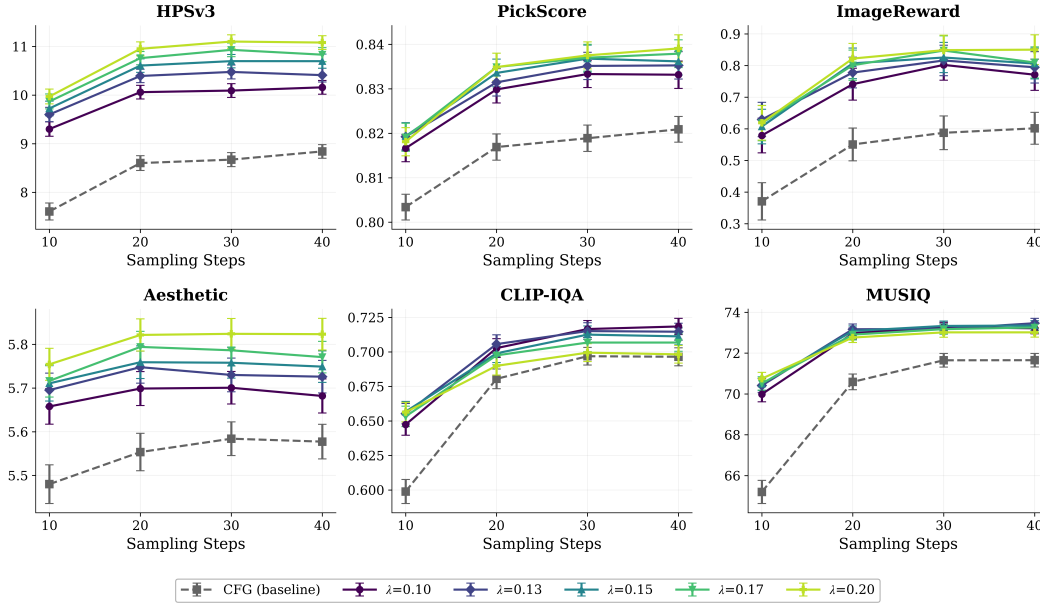


Figure 12: **Effect of sampling budgets and guidance strengths.** Panels evaluate distinct metrics against the total number of generation steps. We compare the standard CFG baseline with RGS at various uniform guidance scales α (labeled as λ). Target rewards scale positively with guidance intensity, while no-reference perceptual quality consistently outperforms the CFG baseline, plateauing at higher guidance levels.

E.5 ROBUSTNESS TO THE ODE SOLVER

The primary results in the main text are based on first-order Euler integration of the generative flow ODE (Lipman et al., 2023). To verify whether RGS generalizes beyond this specific discretization, we evaluate the RGS protocol detailed in Appendix B using four additional classical numerical integrators (Hairer et al., 1993). These include the two-step Adams–Bashforth (AB2) method, which is a multistep method requiring a single model evaluation per step analogous to Euler; the explicit midpoint method; Heun’s method (a trapezoidal predictor–corrector); and the classical fourth-order Runge–Kutta (RK4) method. RGS composes with higher-order solvers in the same way as with Euler: the reward gradient is computed once at the beginning of each step, the solver’s internal velocity evaluations use standard CFG forwards, and the reward-gradient correction from Sec. 3.3 is applied to the solver output afterwards. Its magnitude is scaled relative to the step displacement, $\alpha_i \cdot \text{RMS}(\Delta z)$, with no hyperparameter re-tuning. For these experiments, we implement the three-band RGS schedule from Sec. 3.3. The guidance strengths are set to $\alpha_{\text{early}}=0.30$ for the high-noise phase ($\sigma>0.8$) and $\alpha_{\text{mid}} \in \{0.05, 0.10\}$ for the mid-range phase ($\sigma>0.2$). The latter encompasses both the standard deployed configuration and an augmented mid-band variant, while guidance is omitted in the low-noise tail. The evaluation is conducted on a subset of 250 prompts from the balanced benchmark, with two seeds per prompt. Generations are performed at 1024^2 resolution over 42 steps with a CFG scale of 4.5 using single-head RGS. Each solver’s respective CFG baseline and the two RGS configurations are evaluated on matched prompt–seed pairs. Because these runs use a 250-prompt subset, the baseline CFG numbers deviate slightly from those in Table 3, which uses the full 400-prompt set.

The findings are summarized as follows:

(1) RGS is robust across solvers. As Table 11 shows, RGS reproduces the gains seen with Euler under every integrator: paired per-image HPSv3 rises by roughly two points over the corresponding same-solver CFG baseline, with consistent improvements in the other optimized reward metrics and in no-reference quality. The drift attributable to RGS stays controlled relative to each solver’s base-

Table 11: **Performance of RGS across diverse ODE integrators.** The $\text{FID}\downarrow$ metric is calculated relative to the unguided Euler CFG reference set for the standard CFG evaluation rows (quantifying purely integrator-induced drift), and relative to the *corresponding same-solver* CFG reference set for the RGS evaluation rows (isolating the specific distributional drift attributable to RGS). RGS yields consistent improvements across all target reward and perceptual quality metrics under every solver.

Solver	Method	Reward Model $\uparrow$			Quality / Mode drift		
		HPSv3	ImgReward	PickScore	FID $\downarrow$	MUSIQ $\uparrow$	CLIP-IQA $\uparrow$
(NFE)	(α_{mid})						
Euler (1)	CFG	8.33	0.977	0.870	0.0	73.0	0.697
	+Reward Register (0.05)	10.34	1.167	0.888	64.8	74.9	0.730
	+Reward Register (0.10)	10.44	1.154	0.888	69.3	74.2	0.712
AB2 (1)	CFG	8.29	0.995	0.869	40.1	72.8	0.686
	+Reward Register (0.05)	10.31	1.167	0.887	66.5	74.8	0.731
	+Reward Register (0.10)	10.37	1.160	0.887	72.2	74.2	0.716
Midpoint (2)	CFG	8.35	0.996	0.870	45.0	72.7	0.681
	+Reward Register (0.05)	10.14	1.167	0.887	63.4	74.7	0.728
	+Reward Register (0.10)	10.25	1.155	0.886	69.1	74.2	0.715
Heun (2)	CFG	8.27	1.008	0.869	42.6	72.8	0.687
	+Reward Register (0.05)	10.11	1.167	0.887	64.2	74.6	0.727
	+Reward Register (0.10)	10.25	1.155	0.886	69.7	74.1	0.716
RK4 (4)	CFG	8.38	1.006	0.871	46.8	72.9	0.685
	+Reward Register (0.05)	10.17	1.170	0.888	63.8	74.6	0.728
	+Reward Register (0.10)	10.27	1.152	0.887	70.0	74.0	0.715

line, so the magnitude-matched correction generalizes across integrators without per-solver tuning or any compromise in no-reference quality.

(2) Cross-solver FID mostly reflects trajectory decorrelation, not generative degradation.

Changing the integrator introduces an FID of 40–47 relative to the Euler CFG baseline while the other six metrics stay similar. For a fixed latent seed, different solvers converge to distinct but qualitatively comparable endpoints, so the drift induced by guidance must be measured against the corresponding solver’s own baseline, as in Table 11.

(3) Higher-order solvers attenuate the reward gain only marginally. Midpoint, Heun, and RK4 yield HPSv3 gains of +1.79 to +1.98, versus +2.01 to +2.11 for Euler and AB2. We hypothesize that applying the reward displacement at the end of each step exposes it to the internal evaluations of higher-order solvers, which partially undo the off-manifold correction during the next integration step. The attenuation is only 5–10% of the total reward gain and does not change the qualitative conclusions. Since the single-evaluation AB2 solver closely matches Euler, both offer the best reward–quality trade-off among the integrators tested.

E.6 QUALITATIVE COMPARISON ACROSS PROMPTS

We additionally test RGS under a flat strength $\alpha=0.15$, complementing the deployed schedule in Table 7. Figure 13 uses a matched prompt and seed to isolate the effect of guidance strength; increasing α progressively instantiates stronger preference-aligned details.

F FUTURE WORK

Our results show that reward-gradient guidance applied to noisy latents under a frozen generator improves preference alignment while keeping no-reference quality intact. Two directions seem especially promising for future exploration:

(1) Few-Step Architectures and Video Generation. A natural next step is to evaluate latent reward registers on few-step distilled samplers and video diffusion models. For video generation, the

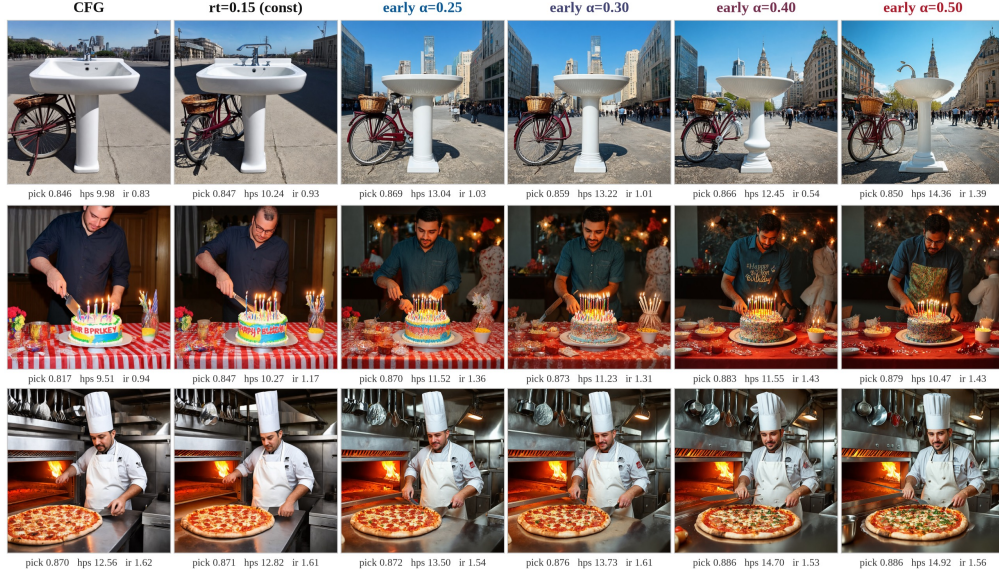


Figure 13: **Qualitative impact of guidance strength variation.** This progression illustrates generated samples subject to monotonically increasing guidance strengths (α), synthesized using an identical text prompt and initial latent seed. The visual sequence demonstrates the progressive instantiation and refinement of preference-aligned details in correlation with the applied guidance magnitude.

register mechanism will require structural adaptations to enforce temporal consistency and coherent motion dynamics across high-dimensional latent sequences.

(2) From RLHF to RLVR. The proposed framework is not inherently limited to subjective human preferences. We plan to explore its integration with RLVR, where the same non-intrusive latent readout could enforce objective criteria—ranging from high-level factual fidelity and spatial reasoning to low-level physical plausibility and strict structural constraints typically found in traditional inverse reconstruction problems.